

Foundational Research & Advances in Quantum Teleportation Protocol

Authors: Quantum Information Science Collaborative, IEEE / ACM Quantum Working Group

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Quantum Teleportation Protocol in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Faxing a quantum particle: you scan an unknown particle with one half of an entangled pair, which destroys the original and produces 2 numbers. When you send those 2 numbers to Bob, his half turns into the exact original particle!"

3. Mathematical Formulation & Unitary Rule

$$|\psi\rangle_C |\Phi^+\rangle_{AB} = \frac{1}{2} \sum_{x,z \in \{0,1\}} |\Phi_{xz}\rangle_{AB} |x\rangle_C$$

Joint Bell expansion reveals that Bob holds the target state up to known single-qubit Pauli rotations X and Z.

4. Step-by-Step Circuit Sequence

Bell state preparation on (1, 2); CNOT(0, 1); H(0); Measure 0 and 1; Classical controlled X and Z on qubit 2.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(3, 3)
# State to teleport on qubit 0 (e.g. rotated state)
qc.rx(1.23, 0)

# Step 1: Create Bell pair on qubits 1 and 2
qc.h(1)
qc.cx(1, 2)
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Cannot transfer continuous complex Transfers exact continuous state using 1 bit (entangled pair) + 2 class

7. Key Applications & Future Research Directions

- Quantum repeater networks for long-distance quantum communication
- Distributed quantum computing and blind quantum computing
- Quantum gate teleportation in fault-tolerant architectures

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Cryptography pipelines.